

KIPRE STAFF TRAINING NEEDS ASSESSMENT (TNA) REPORT

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For

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2026-06-05

Executive Summary

Background

The primary purpose of this Training Needs Analysis (TNA) is to systematically assess institutional competency gaps and evaluate workforce preparedness. This evidence-based approach is designed to inform training priorities and guide strategic human capital development across the KIPRE. The assessment was conducted using a multi-dimensional questionnaire structured into specific modules to capture a holistic view of the workforce. These sections included:

- Basic Information (Directorate, Age, and Sex of the employee)
- Job Role and Responsibilities
- Self Performances Assessing Questions
- Skills and Copetency Assessment
- Training History
- Identification of Training Needs
- Training Delivery Preferences
- Organizational Support
- Future Skills and Emerging Needs
- Additional Comments

By triangulating data from these domains, the analysis identifies not only current skill deficiencies but also the organizational support structures and delivery formats required to bridge them effectively.

Methodology

The assessment was digitized via ODK Central and deployed from 20th – 28th April of 2026. To ensure we captured a representative voice, we used a multi-channel approach—utilizing official email and professional WhatsApp groups. Participation was voluntary, positioned as a collaborative effort to co-design the organization’s future training roadmap. Data Processing & Analysis Once the collection window closed, raw data was ingested into R (v4.5.2). The analytical pipeline focused on cleaning, recoding qualitative responses for statistical depth, and aggregating the results to provide a granular view of needs across all Directorates and Divisions.

Workforce profile

A total of 80 employees, representing 42% of the 189 staff members currently at KIPRE, participated in this assessment. This respondent base spans 23 distinct divisions across seven primary Directorates, ensuring the training roadmap reflects both the core scientific mandate and the essential administrative backbone of the institute. The distribution reveals a workforce concentrated within key technical hubs, most notably the Animal Sciences Welfare and Ethics and Research and Product Development Directorates. In the research domain, the Kenya Snakebite Research and Intervention Division and the Welfare and Ethics Division emerged as the most represented units. This engagement from frontline technical staff—supported by a demographic where 51.2% of respondents are aged between 25 and 44—provides a robust foundation for identifying gaps in laboratory biosafety and modern research protocols. While the gender distribution is

relatively balanced (45% female, 55% male), distinct generational trends are evident: the female cadre is most concentrated in the 35–44 age bracket (36.1%), whereas the male workforce peaks in the 45–54 bracket (34.1%). Administrative and strategic functions are equally well-mapped, with consistent representation from Corporate Services and the Office of the Director General. Notably, only 10% of respondents are aged 55 or older, indicating a predominantly mid-career workforce with significant long-term growth potential. This demographic profile highlights a strategic opportunity to implement cross-cutting training programmes that standardise administrative excellence and ensure the institution is prepared for future succession and evolving digital demands.

Table 1: Workforce distribution by directorate and division

Division / Section	Staff Count
Aswed	
Kenya Snakebite Research And Intervention Division	1
Laboratory Services Section	1
National Laboratory Animal Breeding Division	4
Veterinary Clinical And Diagnostics Services Division	9
Welfare And Ethics Division	11
Cbpgmd	
Human Resource And Administration Division	1
Partnerships And Grants Management Division	3
Csd	
Corporate Communication Division	4
Finance And Accounts Division	4
Human Resource And Administration Division	5
Information Communication Technology Division	1
Planning And Strategy Division	1
Cslsd	
Infectious Diseases And One Health Division	1
Iarad	
Human Resource And Administration Division	1
Odg	
Human Resource And Administration Division	1
Planning And Strategy Division	1
Rpdd	
Data Science And Analytics Section	1
Evolutionary Ecology Conservation And Ecosystem Health Division	7
Infectious Diseases And One Health Division	7
Kenya Snakebite Research And Intervention Division	10
Laboratory Services Section	1
Non Communicable Diseases Division	1
Reproductive Health And Biology Division	4

Table 2: Gender distribution across age brackets

Gender	Under 25 years	25-34 years	35-44 years	45-54 years	55+ years	Total
female	8.3% (3)	22.2% (8)	36.1% (13)	27.8% (10)	5.6% (2)	100.0% (36)
male	6.8% (3)	29.5% (13)	15.9% (7)	34.1% (15)	13.6% (6)	100.0% (44)
Total	7.5% (6)	26.2% (21)	25.0% (20)	31.2% (25)	10.0% (8)	100.0% (80)

Table 3: Job grade distribution by length of service

Job_Grade	Less Than 1 Year	1 3 Years	4 7 Years	8 15 Years	16+ Years	Total
NM1	0	1	0	0	0	1
NM2	0	1	0	0	0	1
NM5	0	0	0	1	3	4
NM6	0	0	0	0	2	2
NM7	0	3	2	2	3	10
NM8	4	3	9	3	4	23
NM9	0	0	0	1	1	2
NM10	0	1	2	4	5	12
NM11	0	0	1	1	8	10
NM12	2	5	1	1	2	11
NM13	0	2	0	1	1	4

1. Which critical competency gaps exist across directorates, divisions, and job grades, and which areas require immediate training intervention?

This question helps management identify

- High-priority skill deficiencies
- Departments with the greatest capacity gaps
- Staff categories requiring urgent support

Likely outputs

- Competency gap heatmaps
- Directorate ranking tables
- Priority training matrix
- Skill deficiency frequencies

Likely recommendations

- Targeted capacity-building programmes
- Directorate-specific training plans
- Prioritised budget allocation

The baseline institutional data establishes that while the organization operates from a fundamentally stable footing, it faces a pervasive mid-level capability ceiling. Across all evaluated competencies, the workforce

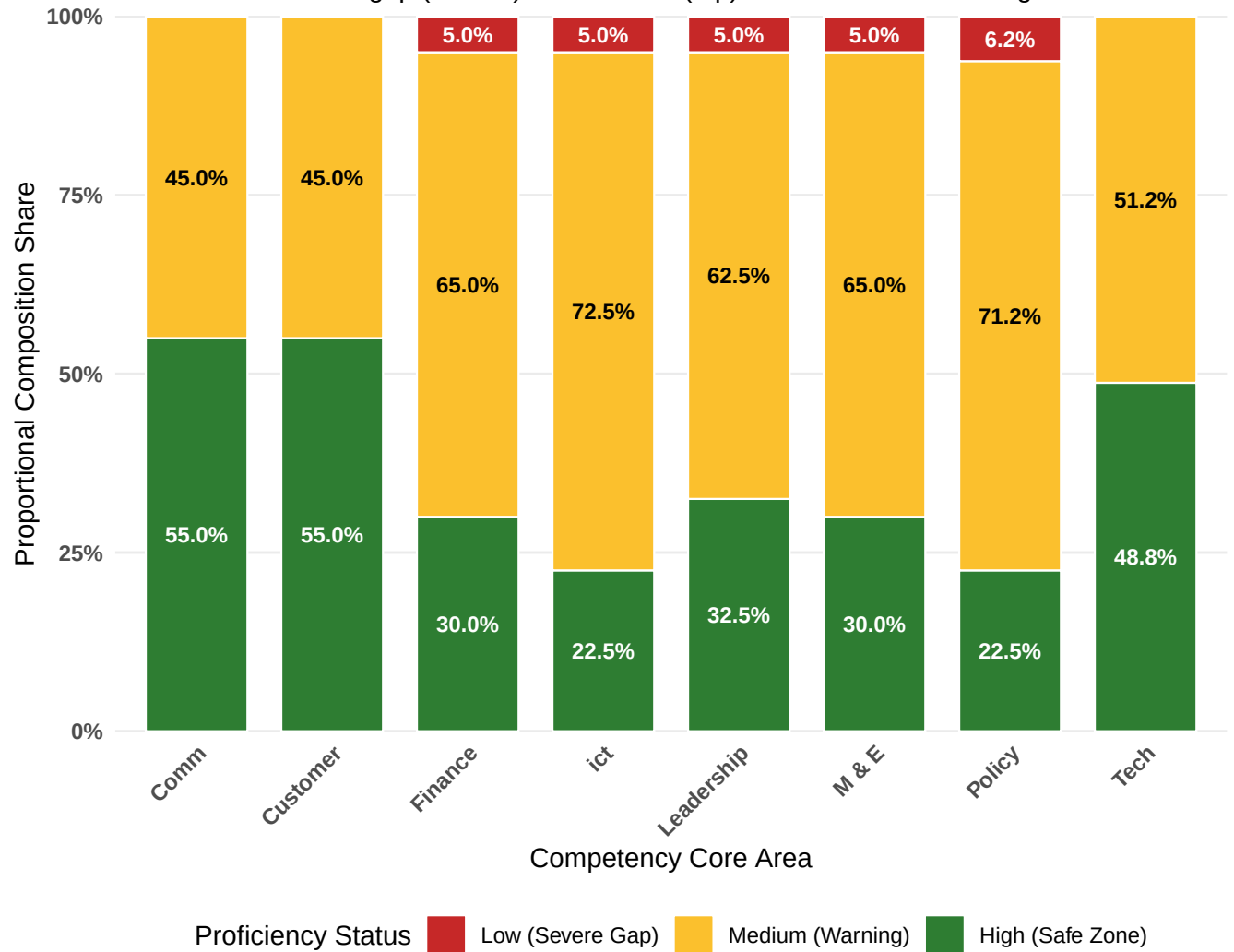
Table 4: Overall Institutional Competency Gap Profile

Average Severity Score	High Proficiency (%)	Medium Proficiency (%)	Low Proficiency (%)
0.66	37.00	59.70	3.30

maintains a baseline Average Severity Score of 0.662, with a healthy 37.0% of responses indicating High Proficiency (Safe Zone) and only a minimal 3.3% falling into the Low Proficiency (Critical Alert Zone). However, a deeper look into the specific competency areas reveals that this stability is heavily anchored by general operations, specifically Comm (55.0% High), Customer (55.0% High), and baseline Tech applications (48.8% High). When these general strengths are set aside, a critical structural vulnerability emerges: a massive 59.7% company-wide majority of competencies sit clustered within the Medium Proficiency (Warning Zone). This exposure is most acute within the organization’s specialized technical and regulatory pillars—specifically ICT (Information and Communications Technology) and Policy implementation—where 72.5% and 71.2% of the workforce, respectively, reside in this warning track. This profile diagnoses an organization whose staff possess excellent general baseline skills, but encounter a clear operational ceiling when handling advanced ICT core infrastructure and specialized Policy requirements. Rather than a widespread institutional crisis, the data uncovers a highly focused training mandate: strategic development resources must pivot away from generic corporate courses, and instead be aggressively targeted toward formalizing internal capacity-building pipelines for advanced ICT systems management, Monitoring & Evaluation (M&E), and Policy Implementation.

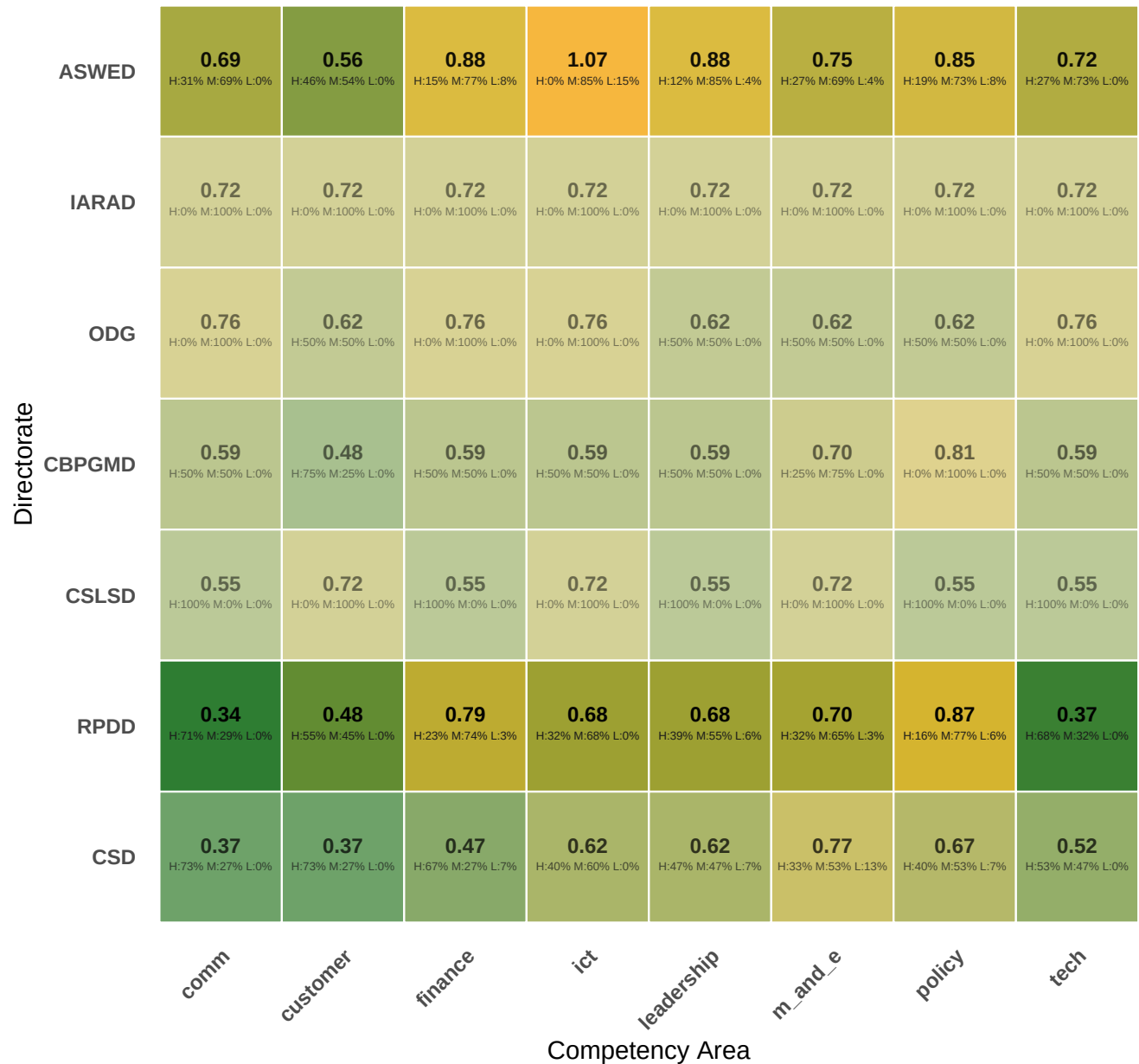
Overall Competency Gap Composition Profile

Stacked from severe gap (bottom) to safe zone (top) with corrected math alignments



Competency Gap Heatmap by Directorate

Shrinkage adjusted metrics (Red indicates highest deficiency severity)



Severity Index



1 (Medium)

Values: High=0, Medium=1, Low=2. Smaller teams are safely shrunk toward global averages to eliminate sample bias.

Competency Gap Heatmap by Division

Shrinkage adjusted metrics (Warm tones indicate higher training needs)



Severity Index

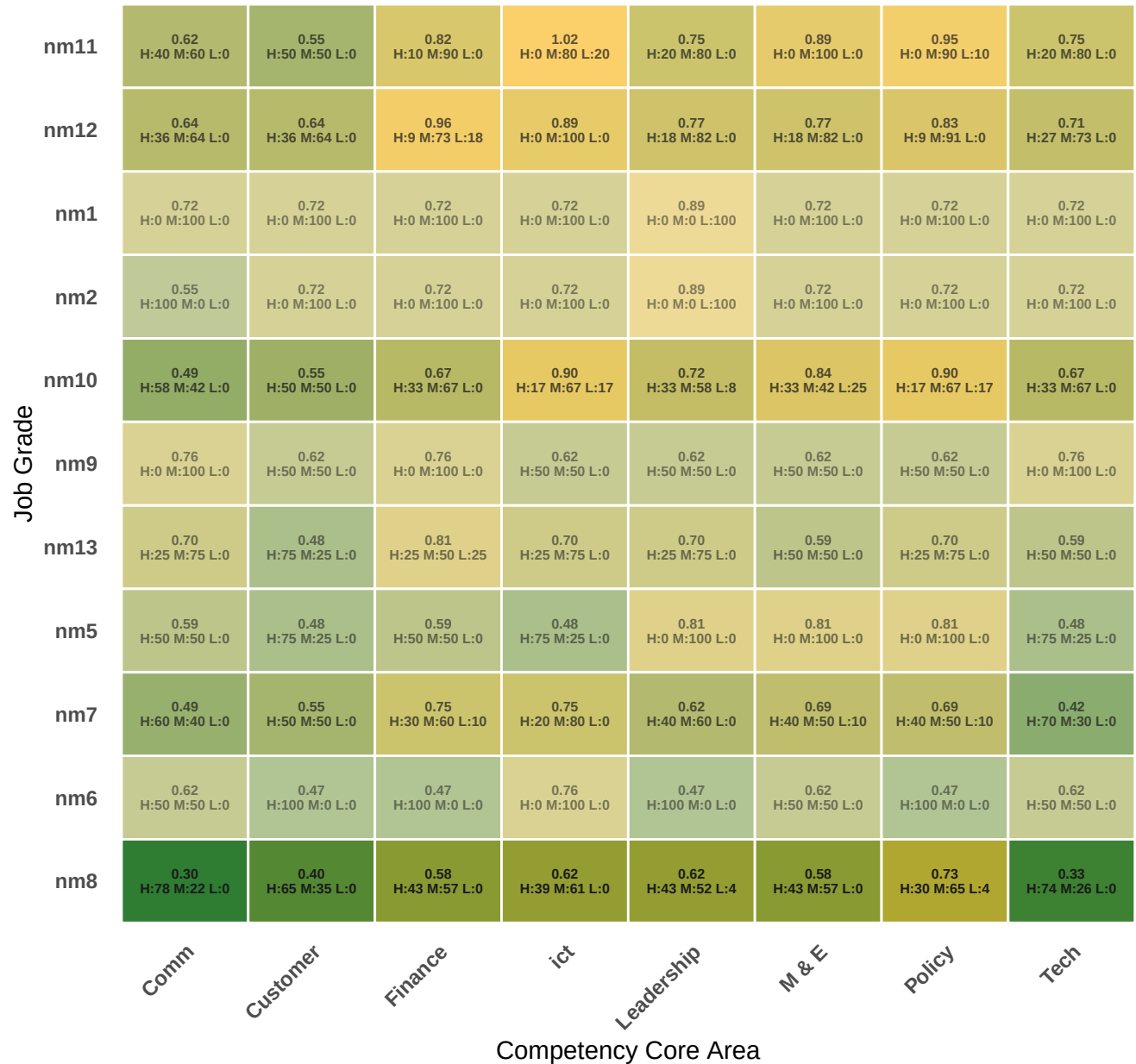


1 (Medium)

Labels show: Adjusted Score (Top) / Employee Skill Breakdowns (Bottom).

Competency Gap Heatmap by Job Grade

Shrinkage adjusted metrics (Warm tones indicate highest deficiency severity)



Severity Index



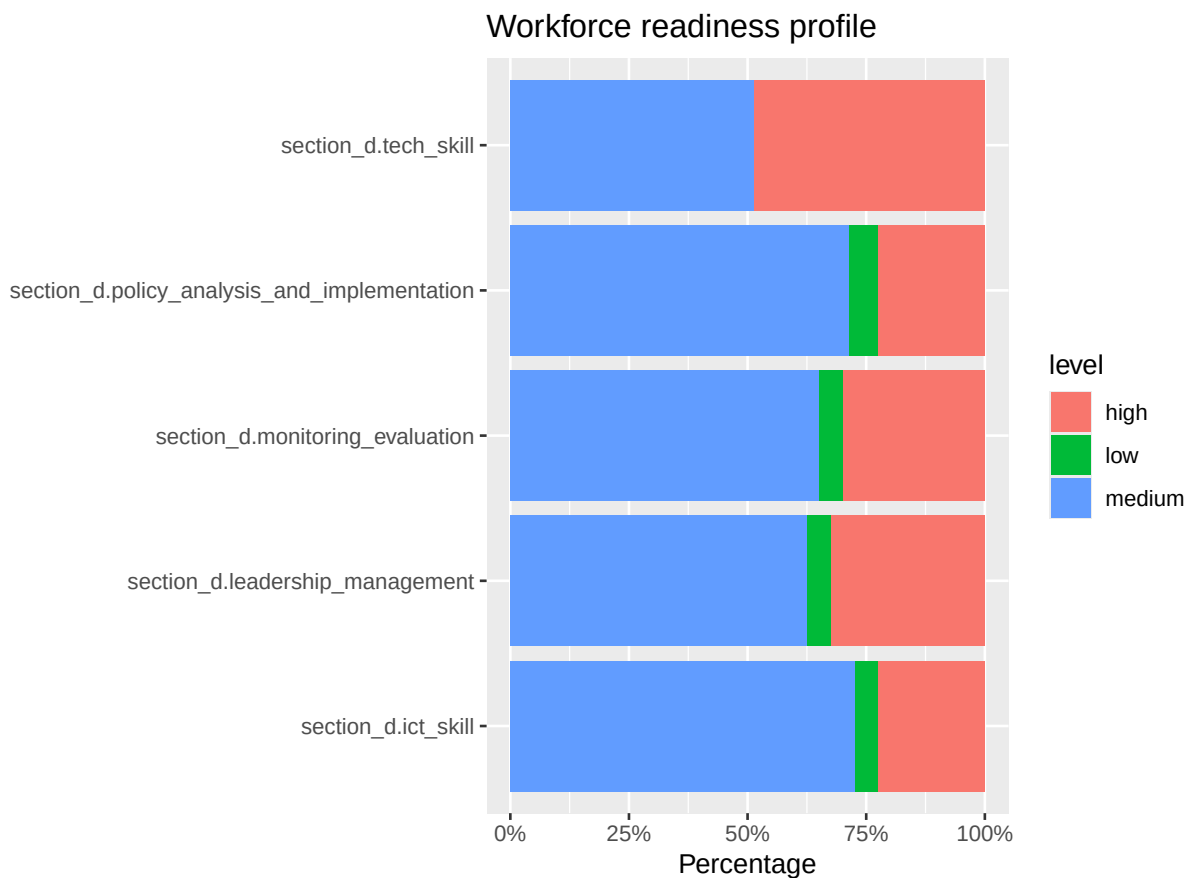
1 (Medium)

Values: High=0, Medium=1, Low=2. Labels display: Shrunk Score / raw H-M-L percentages.

2. To what extent are employees adequately prepared to meet emerging organizational, technological, and policy-related demands?

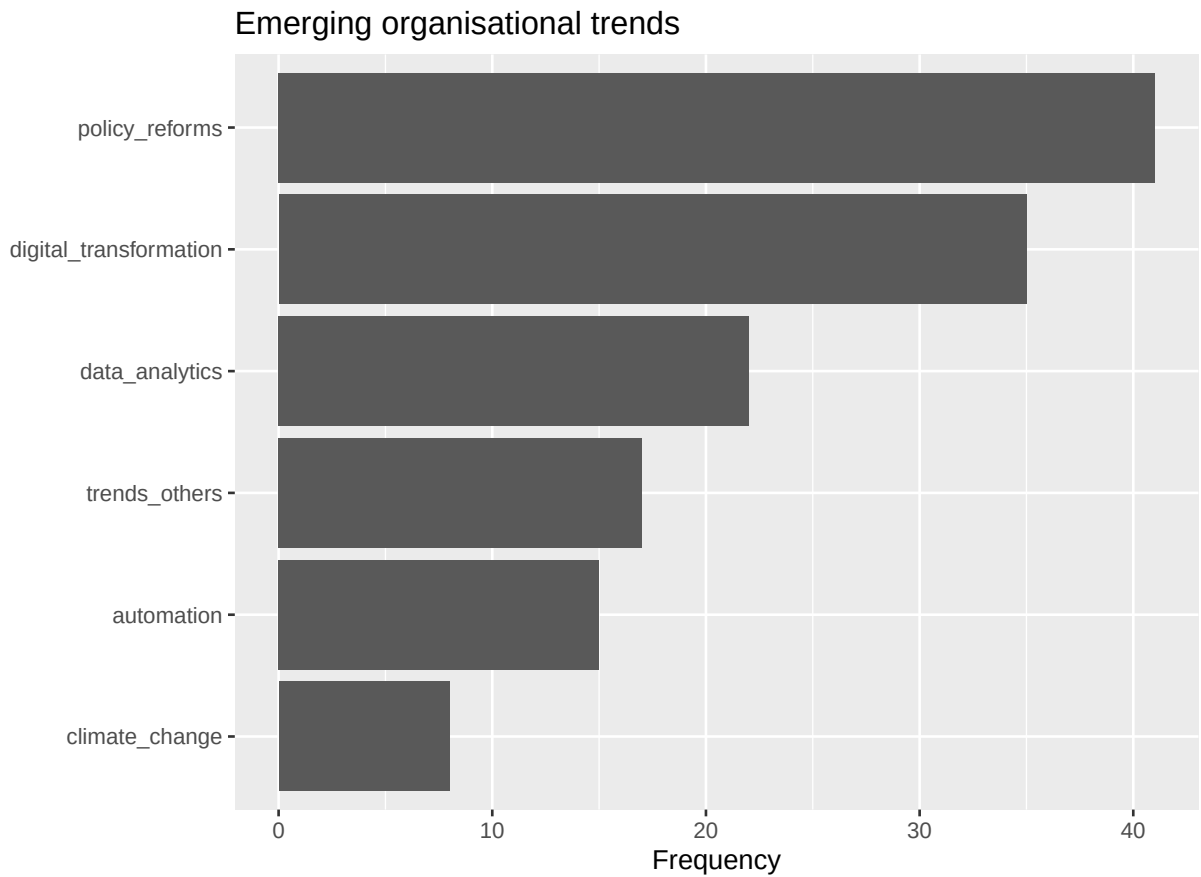
```
## # A tibble: 1 x 4
##   Mean    SD   Min   Max
##   <dbl> <dbl> <dbl> <dbl>
## 1  2.27 0.353   1.4     3
```

```
## # A tibble: 7 x 3
##   section_a.directorate readiness    n
##   <chr>                <dbl> <int>
## 1 CSLSD                2.6     1
## 2 CSD                  2.37    15
## 3 CBPGMD               2.35     4
## 4 RPDD                 2.34    31
## 5 ODG                  2.3     2
## 6 ASWED                2.11    26
## 7 IARAD                 2     1
```



```
## # A tibble: 6 x 2
```

```
## section_i.emerging_trends      n
## <chr>                          <int>
## 1 policy_reforms               41
## 2 digital_transformation       35
## 3 data_analytics               22
## 4 trends_others                17
## 5 automation                  15
## 6 climate_change               8
```



```
## # A tibble: 211 x 2
##   skill                                n
##   <chr>                             <int>
## 1 Management in Tourism              6
## 2 Leadership skills                  5
## 3 Communication skills               3
## 4 Data analytics                     3
## 5 Animal husbandry                  2
## 6 Equipped with management skills     2
## 7 Leadership                         2
## 8 Leadership management               2
```

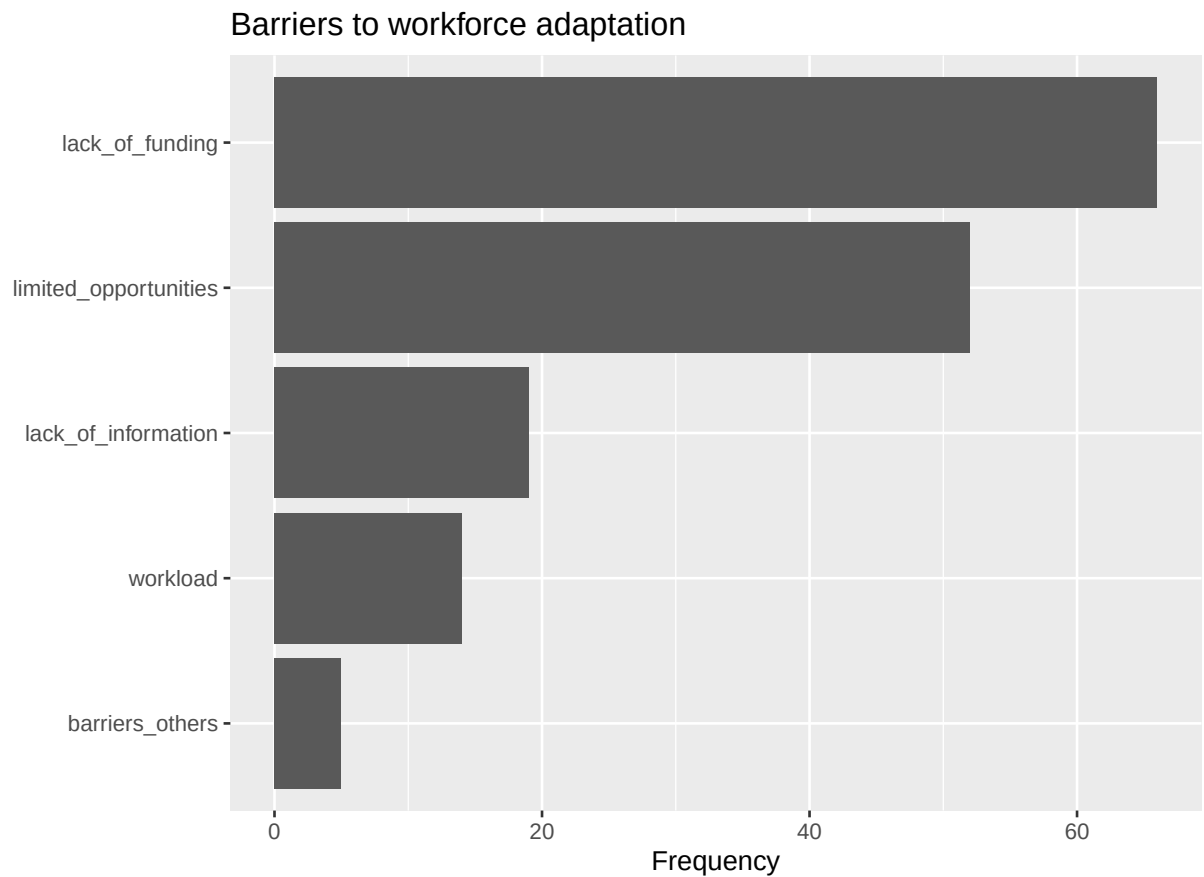
```
## 9 100% fluency in use of automated Finance System(ERP) 1
## 10 A Degree in Management or Guidance and counselling 1
## # i 201 more rows
```

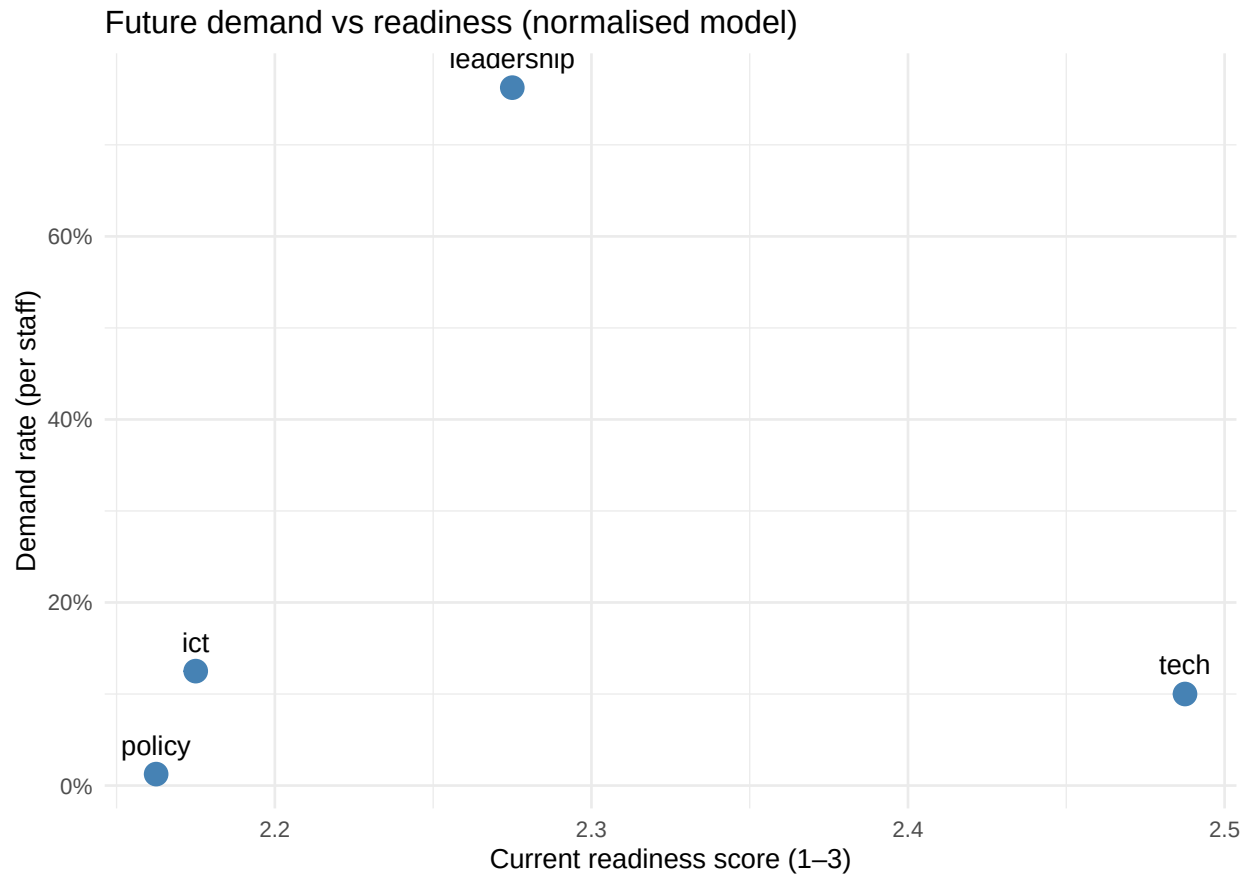
Translational research and innovation skills to bridge laboratory into clinical practice
 Use of AI and emerging technologies in disease diagnosis
 Equipped with the necessary skills to design laboratory infrastructure
 Technical skills in new diagnostic technologies
 Antibody engineering—I am interested in learning about designing of poly-specific nanobody libraries, phage display and recombinant protein engineering
 Scientific writing and communication for higher education
 Research and data management skills in grant writing & funding acquisition
 Research and development skills in vaccine development
 Quality assurance, I will need knowledge of quality management systems, including standards like ISO, to ensure accuracy and reliability
 Project management and leadership skills to coordinate teams, manage research projects, and ensure timely completion
 Positive Reinforcement Training (PRT)
 Modern laboratory equipment, I will need the ability to operate, calibrate, and maintain modern automated analyzers and machines
 Monitoring and evaluation frameworks for assessing conservation status
 Mastering the skills in molecular biology and procedures applied in molecular laboratories
 Machine learning and AI
 Leadership and project management to coordinate research projects
 Laboratory techniques
 Integrated skills with leadership roles that involves multidisciplinary approach both technical and managerial
 ICT skills
 Input screening & automation – I need to learn about robotics for venom fractionation, automated ELISA platforms, and high-throughput screening
 Grant proposal development and resource mobilization
 GIS, remote sensing, and climate modeling
 Firewall and encryption
 Facilitation skills
 Effective field work
 Environmental protection
 Laboratory skills, I will need skills in using laboratory information systems, computer data entry and digital reporting in laboratories
 Data analysis and interpretation, I will need basic skills in analyzing laboratory results using tools like Excel and understanding of statistical software
 Data analysis and interpretation
 Continuous development veterinary and animal health
 Colony Genetics and Advanced Breeding
 Climate Change and Environmental Conservation
 On prevention, I will need updated knowledge in biosafety, waste management, and infection control to handle samples safely
 Better understand laboratory procedures
 Be able to do more
 Application of Synthetic Biology
 All skills updated by international standards
 Advanced technical skills in Next-generation sequencing methods—deeper learning and more advanced practical skills
 Laboratory techniques, I will need to improve my technical skills in areas such as microbiology, hematology, and clinical chemistry
 Advanced data analysis
 Advanced causal inference
 100% fluency in Bengali

```
## # A tibble: 8 x 2
##   section_d.improve_skill      n
##   <chr>                    <int>
## 1 technical_professional_skills 57
## 2 leadership_management        56
## 3 policy_analysis_and_implementation 49
## 4 monitoring_and_evaluation     45
## 5 financial_management          43
## 6 ict_skills                   42
## 7 communication_skills         29
## 8 customer_service             16

## # A tibble: 2 x 3
##   section_f.level_training_needs      n      pct
##   <chr>                    <int> <dbl>
## 1 high                      72    90
## 2 medium                    8    10
```

```
## # A tibble: 5 x 2
##   section_h.barrier      n
##   <chr>              <int>
## 1 lack_of_funding      66
## 2 limited_opportunities 52
## 3 lack_of_information   19
## 4 workload              14
## 5 barriers_others        5
```





- Future workforce readiness
- Digital transformation preparedness
- Emerging competency requirements
- Institutional adaptability

Likely outputs

- Emerging skills rankings
- Workforce readiness summaries
- Future skills dashboard
- Trend analysis

Likely recommendations

- Future-focused training programmes
- Digital skills initiatives
- Leadership and innovation training
- Strategic workforce development planning

Key recommendations

Priority actions

Conclusion

Strategic implications

7. Exemplary visualisations